

HAT-P-37b

R-2374

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_201005 · created 2026-08-20T00:39:21+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459127.7671 +/- 0.0059 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.134 +/- 0.039
Transit depth (Rp/Rs)^2	1.81 +/- 1.06 [%]
Orbital Inclination (inc)	89.88 +/- 1.87
Airmass coefficient 1 (a1)	0.97 +/- 0.039
Airmass coefficient 2 (a2)	0.0062 +/- 0.0078
Scatter in the residuals of the lightcurve fit is	4.02 %
Best Comparison Star	#2 - [366, 429]
Optimal Aperture	2.94
Optimal Annulus	10.94
Transit Duration (day)	0.1019 +/- 0.0065

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.134 $\pm$ 0.039 vs 0.1394 (deviation 0.11 σ)
Duration fit vs published	0.1019 d vs 0.0983 d (deviation 0.42 σ)
Mid-time O-C	15.3 min
Depth SNR	2.6
Residual scatter	4.02 %

Light curve

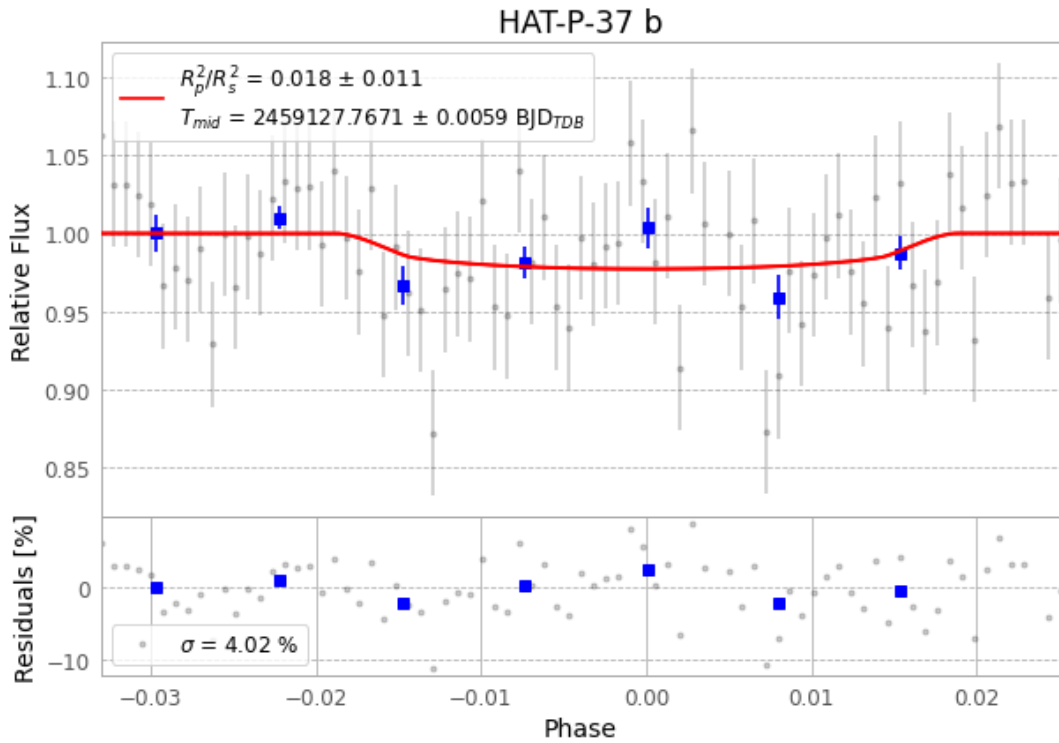


Figure 1 — FinalLightCurve_HAT-P-37 b_2020-10-04.png (SHA-256 fb95cb398ac029ec...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [344, 297], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0ec7444424b14dc4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

79 FITS frames (52 MB), HATP-37201005040912.FITS → HATP-37201005080312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-201005.log (23d8c5d8b6f3...), FinalLightCurve_HAT-P-37 b_2020-10-04.png (fb95cb398ac0...), FinalLightCurve_HAT-P-37 b_2020-10-04.csv (e719d5d37b8c...)

Compute resources

Wall time 107.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 00:39Z.